

Eccentric ruthenium plaque radiotherapy of posterior choroidal melanoma

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Abstract

Background The aim of this work is to report on conservation of visual acuity after eccentric ruthenium¹⁰⁶ (Ru-106) brachytherapy of posterior choroidal melanoma.

Methods Fifty-four patients with choroidal melanoma extending to within 5 mm of the optic disc or fovea were treated at a single institution during the 3 years up to July 31, 2007. The plaque was positioned eccentrically with its posterior edge aligned with the posterior tumor margin to reduce the radiation dose to the optic disc and fovea. The main outcome measures were local tumor control and conservation of vision of 6/12 or better, according to baseline variables.

Results The tumors had a mean longest basal diameter of 12.0 mm and a mean thickness of 3.1 mm. The posterior tumor margin extended to within 3 mm of the fovea in 30 patients (56%). In the 24 eyes with the posterior tumor margin located 3.1–5.0 mm from the fovea, 74.9% retained visual acuity of 6/12 or better 4 years after treatment (95% confidence interval [CI], 55.5–94.3%). In the 30 eyes with tumor extension to within 3.0 mm of the fovea, then 3 years after treatment only 25.3% retained such vision (95% CI, 5.3–45.3%). Recurrence at the posterior tumor margin occurred in two patients.

Conclusions Eccentric ruthenium¹⁰⁶ plaque radiotherapy of posterior choroidal melanoma achieves good rates of local tumor control and conservation of vision if special measures are taken to ensure that the plaque is positioned correctly.

Keywords Plaque brachytherapy · Ruthenium¹⁰⁶ · Posterior choroidal melanoma · Eccentric placement

Introduction

Most choroidal melanomas are now treated by episcleral plaque radiotherapy with the aim of preventing metastatic spread while conserving the eye and vision. Brachytherapy with Ru-106/Rh-106 applicators was introduced for the treatment of posterior uveal melanomas in 1964 by Lommatzsch and Vollmar [1]. Many studies have since been published on this form of radiotherapy, reporting on outcomes such as patient survival, local tumor control, as well as ocular and visual conservation.

To avoid marginal tumor recurrence, the conventional practice is to center the plaque over the tumor base so that it overlaps all the visible tumor margins by at least 2 mm [2, 3]; however, this causes visual loss if the tumor extends close to the optic disc or fovea. Since 1988 (and more consistently since 1993) one of the authors (B.D.) has treated posterior choroidal melanomas by placing the plaque eccentrically in relation to the tumor so that it is as far as possible from the optic nerve and fovea, thereby reducing the risks visual loss. Outcomes of patients treated with this approach have been reported previously, but only as part of a larger series of patients with choroidal melanoma, many of whom were not treated with an eccentric plaque [4, 5].

The main aim of this study is to document visual outcomes in patients with choroidal melanoma extending to within

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The authors have full control of all primary data and agree to allow Graefes' Archive for Clinical and Experimental Ophthalmology to review data if requested.

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5 mm of the optic disc or fovea and treated with an eccentrically positioned Ru-106 plaque.

Materials and methods

Patients were included in this study if treated for choroidal melanoma during the 3 years up to the July 31, 2007. They were excluded if they were treated for a non-melanomatous tumor, the tumor had previously been treated by other another method, or if the tumor did not extend to within 5.0 mm of the optic disc or fovea.

In keeping with conventional practice, the diagnosis of melanoma was based on the following features: tumor thickness exceeding 2 mm, confluent orange pigment on the tumor surface, serous retinal detachment, symptoms, and documented tumor growth. The likelihood of melanoma was assessed according to the number and severity of these features. Diagnostic biopsy was rarely needed.

The distance between the posterior tumor margin and the center of the optic disc was measured pre-operatively by ultrasonography, whereas the distance from the tumor to the fovea was determined by reviewing fundus photographs, assuming the optic disc to have a horizontal diameter of 1.5 mm and then using this as a reference.

As a rule, brachytherapy using a 15-, 20-, or 25-mm ^{106}Ru plaque (Eckert & Ziegler BEBIG, Berlin, Germany) was our first choice of treatment for choroidal melanoma unless the tumor height exceeded 5.4 mm or if the posterior tumor margin extended to within 2 mm of the optic disc, so that risk of optic neuropathy was considered high. Preoperative and follow-up assessments were performed as previously described [6]. Follow-up assessments were alternated between our center and the referring ophthalmologist until the probability of complications was considered low (i.e., < 1%), when patients were discharged to their ophthalmologist for annual review. If no follow-up information was automatically received from the referring ophthalmologist, it was requested by mail.

Our surgical techniques have been previously described [4]. Briefly, to ensure accurate positioning of the plaque in relation to the tumor, any overlying extraocular muscles were routinely disinserted. Before muscle disinsertion, sutures were placed in the muscles and the knot-to-limbus distances were measured and recorded. The anterior and lateral tumor margins were identified by transpupillary transillumination or indentation. All tumors were treated with the posterior edge of the plaque aligned to the posterior tumor margin. This was achieved by marking the sclera with a pen at a point anterior to the tumor corresponding to the difference between the ultrasound longitudinal basal tumor dimension and the plaque diameter. The anterior margin of a transparent plastic template was sutured to the sclera with releasable knots so that its anterior edge was aligned with the ink mark on the sclera. Correct positioning

was confirmed by inserting the tip of a right-angled, 20-G transilluminator into perforations in the template and performing binocular indirect ophthalmoscopy. The plaque was considered to be in good position if the light from the transilluminator was located at the posterior tumor margin (i.e., “sunset sign”). The template was then replaced with the radioactive plaque and a mattress suture was placed over the plaque to ensure that it was firmly applied to the globe. Finally, any disinserted muscles were returned to their correct anatomic position, ensuring that the knot-to-limbus distances were the same as those measured before disinserting the muscle.

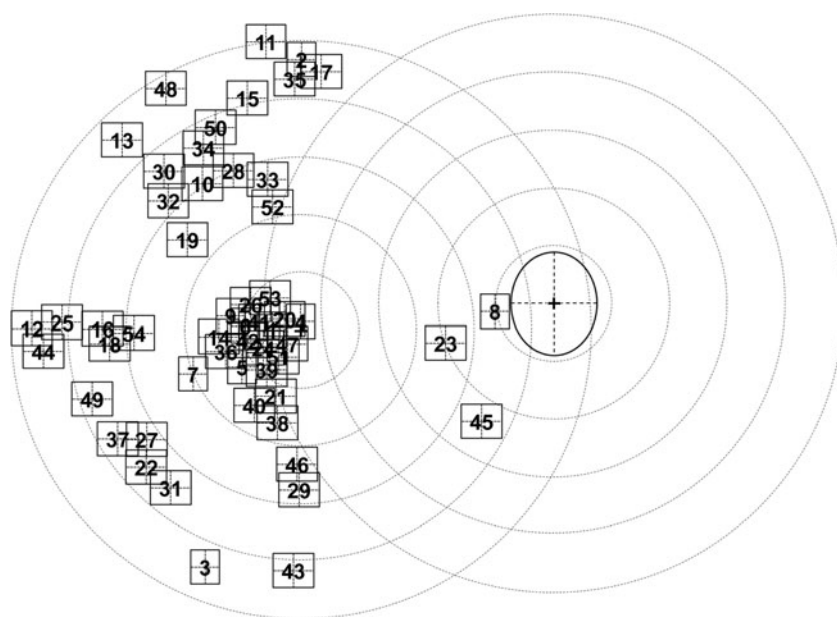
Six months postoperatively, the distribution of the radiation-induced choroidal atrophy was assessed by indirect ophthalmoscopy and if there was any suspicion that part of the tumor had not received sufficient radiation, this area was treated with transpupillary thermotherapy. A minimum scleral dose of 300–350 Gy and a minimum apex dose of 80–100 Gy were prescribed, using the manufacturer’s ASMW specifications (ASMW = standardization office of the former German Democratic Republic). At our center, visual acuity was measured using a LogMAR chart, with the patient’s usual spectacle correction and a pinhole if necessary. If the patient was unable to count fingers centrally, then the test was repeated with the examiner’s hand in each quadrant. Local tumor recurrence was diagnosed if unequivocal expansion of any tumor margin was seen when comparing the ophthalmoscopic appearances with previous color photographs or if ultrasonography indicated that the tumor thickness had increased by more than 0.5 mm.

The tenets of the Declaration of Helsinki were followed. Informed consent for eccentric plaque radiotherapy was obtained routinely together with generic signed consent for using clinical data, images and tissues for research, teaching, and audit. Since this was an audit, ethical committee approval was not required.

Results

The 54 patients comprised 16 men (30%) and 38 women (70%) with a mean age of 65.9 years at the time of treatment (range, 29–88 years). The affected eye was the left eye in 26 patients (48%) and the right eye in 28 (52%). The pre-operative visual acuity was 6/12 or better in 41 patients (76%). The tumors had a mean basal diameter of 12.0 mm (range, 6.4–17.8 mm) and a mean thickness of 3.1 mm (range, 0.8–5.4 mm). In relation to the fovea, the tumor was located: superiorly (3); super-nasally (1); infero-nasally (2); inferiorly (3); infero-temporally (13); temporally (18); and supero-temporally (14) (Fig. 1). The posterior tumor margin extended within 3.0 mm: of optic disc and fovea in seven patients (13%); fovea only in 23 (43%); disc only in two (4%); and neither in 22 (41%).

Fig. 1 Diagram showing approximate quadrant and distance between posterior tumor margin and fovea



The plaque diameter was 15 mm in 12 patients, 20 mm in 37 patients, and 25 mm in five patients. The median plaque duration was 52 h (range, 30–214 h). The median scleral dose was 350 Gy (range, 238–931 Gy) and the median basal dose rate was 7.02 Gy/h (range, 3.87–11.92). The median apex radiation dose was 98.5 Gy (range, 64–185) and the median apical dose rate was 1.82 Gy/h (range, 0.37–5.08).

The median time to the last known visual status was 3.5 years (range, 0.07–6.6). There were 49 patients resident in mainland Britain who had a median follow-up time of 4.8 years (range, 0.5–6.9). Eleven of these patients died, the certified cause of death being metastatic disease in five cases.

The last known visual acuity was: 6/12 or better in 30 patients (56%); 6/18–6/60 in 11 (20%); 3–60 count fingers in ten (19%) and hand movements to light perception in three (6%). No eyes were enucleated. Figure 2 shows conservation of visual acuity according to the distance from the posterior tumor margin to the fovea. If this was located 3.1–5.0 mm from the fovea, then 74.9% of eyes retained visual acuity of 6/12 or better 4 years after treatment (95% confidence interval [CI], 55.5–94.3%). If the tumor extended closer than 3.1 mm from the fovea, then 3 years after treatment only 25.3% retained such vision (95% CI, 5.3–45.3%) (Fig. 2).

Four patients developed tumor recurrence. The first case, a 64-year-old man (case 53), developed a posterior, marginal tumor recurrence 19 months after treatment of a temporal choroidal melanoma in the right eye. The tumor had a basal diameter of 12.1 mm and a thickness of 3.2 mm. The patient underwent proton beam radiotherapy. Seventeen months later, the tumor had responded to treatment but the visual acuity had deteriorated to 6/12. The second patient, a 53-year-old man (case 39) developed a posterior marginal recurrence 60

months after treatment of an infero-temporal choroidal melanoma, which extended to within a disc diameter of the fovea and which measured 12.9 mm in largest basal diameter with a thickness of 4.1 mm. He was treated with proton beam radiotherapy. The third case, a 29-year-old woman, developed subconjunctival tumor seeding 6.5 years after plaque radiotherapy with trans-retinal tumor biopsy. She was successfully treated by excision and cryotherapy and was well at the close of the study 3 months later. Her tumor contained epithelioid cells but showed no chromosome 3 loss. One patient, a 67-year-old woman, developed a central recurrence of a right, infero-temporal choroidal melanoma, which increased in thickness from 2.5 to 3.3 mm 6 months after plaque radiotherapy. This was treated with proton beam radiotherapy, with visual acuity of 6/7.5 51 months later. (Fig. 3)

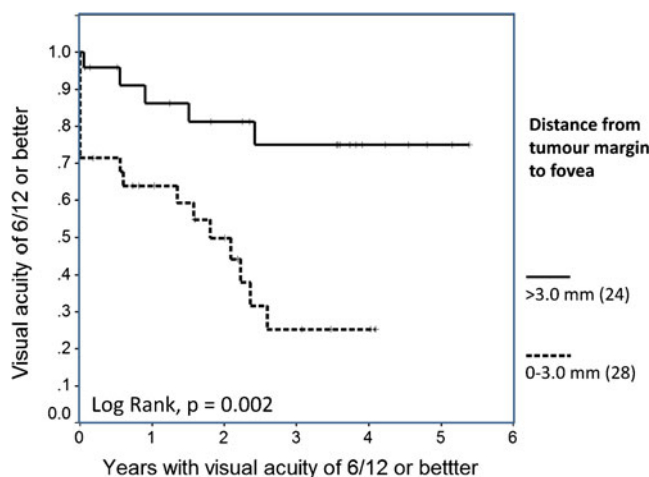


Fig. 2 Kaplan-Meier curves showing conservation of vision of 6/12 or better according to distance between the tumor margin and fovea

We treated three tumors with transpupillary thermotherapy (TTT) because the visible choroidal atrophy did not extend far as far posteriorly as we had intended (Table 1). Two other tumors were treated with TTT for macular edema (Table 1).

Other complications included: cataract in five patients (9.2%); macular edema in five patients (9.2%); radiation maculopathy in four patients (7.4%) (two of four patients got diabetes); macular degeneration in three patients (5.5%); epiretinal membrane in one patient (1.8%); exudative retinal detachment in one patient (1.8%) (the epiretinal membrane should be related with the vitreo-retinal surgery performed 1 year before of our treatment, while the exudative retinal detachment was already seen at diagnosis time); optic neuropathy in one patient (1.8%) diplopia in one patient (1.8%); episcleritis in one patient (1.8%); retinal tear in one patient (1.8%).

Of the 24 patients whose tumor margin was located 3–5 mm from the fovea, six (25%) lost vision of 6/12 or better. The causes of visual loss were pre-existing epiretinal membrane related to retinal detachment surgery preceding the plaque radiotherapy in (one patient); macular edema/retinopathy (four patients, three of whom had diabetes mellitus); and uncertain cause in one patient, who subsequently died of renal failure. Two patients had a bulky tumor (one measuring 12.6 mm basally with a thickness of 5.1 mm and the other measuring 14.6 mm in diameter with a thickness of 4.1 mm).

Discussion

The main finding of this audit is that many patients with posterior tumor extension retained good visual acuity for at least 2 years, despite posterior extension of their tumor. Such patients would inevitably have received higher doses of radiation at the optic disc and fovea had the ruthenium plaque been positioned conventionally, overlapping the entire tumor margin by 2 mm. Despite the limited safety

margin, only two patients developed marginal local tumor recurrence.

To our knowledge, this is the first study specifically evaluating brachytherapy of posterior choroidal melanomas with an eccentrically positioned ruthenium plaque. Other retrospective studies have investigated outcomes after centrally positioned applicators containing gold-198, iodine-125, ruthenium-106, iridium-192 or palladium-103 with varying results [7–27]. The main weakness of our study is the short follow-up. Although visual loss following ruthenium plaque radiotherapy usually occurs within the first two post-operative years, there is scope for long-term outcome studies. Another weakness is that in many patients the latest visual acuity was measured at their local hospital using a Snellen chart; however, most patients lived far from our center and we did not consider it ethical to oblige them to return to our clinic merely for the purposes of this study, in view of the cost and inconvenience they would have incurred. It is possible that the recurrence rate was so low because some of the tumors were benign; however, we consider this to be unlikely because the tumors showed serous retinal detachment and/or orange pigment highly suggestive of malignancy. In the early part of the study, three tumors were treated with delayed TTT because the visible radiation-induced choroidal atrophy did not extend as far posteriorly as we would have liked so that were unsure as to whether the entire tumor had received sufficient radiation. None of these tumors had shown any evidence of growth. We subsequently learned that such TTT was unnecessary and adopted a wait-and-see approach.

Several studies associated posterior tumor extension with visual loss [13, 28–30]. Shields et al. reported tumors less than 5 mm from the optic disc or foveola to have poor visual acuity (20/200 to no light perception) in 35% at 5 years, whereas those 5 mm or more from the optic disc and foveola showed poor visual acuity in 25% at 5 years [31]. Another study performed by Bergman et al. described the posterior tumor border within 2 mm of the foveola in 47%, and within 2 mm of the optic disc in 37%. In their series, 31% of patients had a

Fig. 3 **a** Pre-operative choroidal melanoma. **b**

Post-brachytherapy choroidal atrophy surrounding the tumor indicating adequate positioning of the plaque and central local tumor recurrence

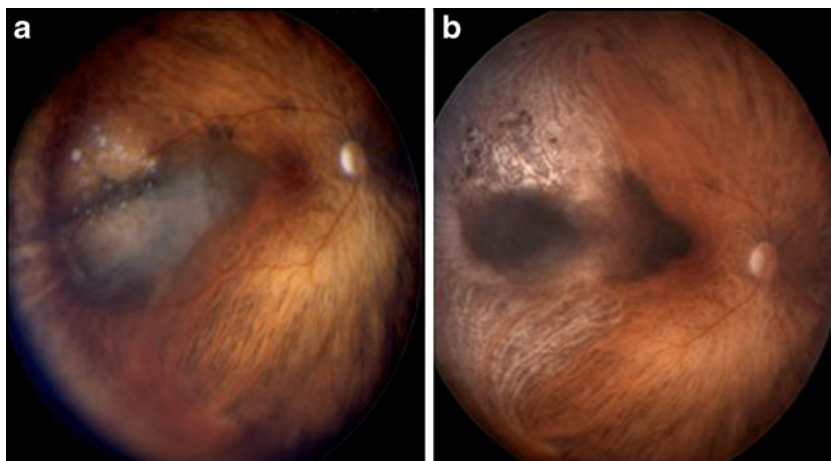


Table 1 Demographics of 54 patients with posterior uveal melanoma treated with 106Ru brachytherapy from July 2004 to July 2007, with characteristics of posterior uveal melanomas and plaque radiotherapy parameters (*CF* counting fingers; *HM* hand movements; *TTT* transpupillary thermotherapy)

Number	Sex	Eye	Initial visual acuity	Quadrant	Large ultrasound diameter (mm)	Ultrasound height (mm)	Distance of tumor from disc (mm)	Distance of tumor from fovea (mm)	Plaque diameter (mm)	Apex dose (Gy)	Final vision	Follow-up time (months)	Recurrence	Adjuvant TTT
1	Female	Left	6/6	Superior	7.4	2.1	4.8	4.71	15	100	6/12	26.00		TTT
2	Female	Left	6/24	Temporal	16	5	4	0.42	25	100	6/60	17.00		
3	Female	Right	6/5	Infero-temporal	12.3	3.4	7.5	4.5	20	100	6/5	6.00		
4	Female	Right	CF	Temporal	11.5	5.2	3.8	0	20	100	HM	82.00	Episcleral recurrence (6.5 years after plaque and transretinal biopsy)	
5	Female	Left	6/9	Infero-temporal	12.8	3.3	6.5	1.28	20	90	6/60	32.00		
6	Female	Left	6/6	Infero-temporal	13.3	4.4	4.7	1	20	90	6/15	32.00		
7	Female	Right	6/9	Infero-temporal	10.3	2.2	6.3	2	20	152	6/15	19.00		
8	Female	Right	6/9	Nasal	12.2	1.6	1.5	4.1	20	114	6/7.5	7.00		
9	Female	Right	6/24	Temporal	9.9	2.6	5.9	1.2	15	100	6/19	30.00		
10	Female	Right	6/9	Supero-temporal	14	2.4	4.9	3	20	104	6/24	28.00		
11	Female	Left	6/9	Supero-temporal	10.9	2.4	4.7	4.98	20	100	6/12	15.00		TTT
12	Female	Right	6/6	Temporal	8.1	2.3	9.7	4.75	15	100	6/9	18.00		
13	Male	Right	6/9	Supero-temporal	14.7	4.7	5.2	2.5	20	80	6/60	19.00		
14	Female	Right	6/12	Temporal	12.3	3.9	5.5	1.41	20	73	6/9.5	44.00		
15	Female	Left	6/6	Supero-temporal	13.7	2.9	6.3	4.29	20	90	6/5	27.00		
16	Female	Left	6/6	Temporal	10.1	4.2	5.9	3.52	20	65	6/24	30.00		
17	Female	Right	6/5	Superior	8	2	4.3	4.5	15	97	6/6	18.00		
18	Female	Left	6/6	Temporal	10.9	2.3	5.8	3.2	20	95	6/5	55.00		
19	Male	Left	6/18	Supero-temporal	10.4	2.5	5.6	2.56	20	80	6/18	5.00		TTT
20	Female	Left	6/60	Supero-temporal	11.6	4.3	2.7	0	20	80	3/60	21.00		
21	Female	Left	6/6	Infero-temporal	14.6	4.1	7.6	3.2	20	80	6/15	12.00		
22	Female	Right	6/5	Infero-temporal	10	2.5	7.5	3.75	20	100	6/7.5	67.00	Central tumor recurrence (6 months after brachytherapy)	
23	Female	Left	6/12	Infero-nasal	13.4	5.9	1.9	2.5	20	64	6/19	10.00		
24	Male	Right	6/18	Temporal	14.2	4.6	4.7	0.47	20	64	6/15	22.00		
25	Female	Right	6/6	Temporal	10.6	0.8	6.2	4.64	20	185	6/7.5	55.00		
26	Female	Left	6/60	Temporal	13.9	2	2.95	0.3	20	174	HM	33.00		
27	Male	Left	6/6	Infero-temporal	9.6	2.8	8.7	3.49	15	121	6/7.5	20.00		
28	Female	Right	6/12	Supero-temporal	11.6	3.7	4.4	3	20	103	6/48	48.00		TTT for macular edema
29	Female	Right	6/24	Inferior	14.8	6.8	3.5	2.8	20	80	6/60	28.00		
30	Male	Left	6/12	Supero-temporal	12.5	4.7	6.8	3.61	20	85	6/19	47.00		
31	Female	Right	6/6	Infero-temporal	10.7	2.6	7.5	3.8	20	125	6/5	26.00		
32	Male	Left	6/5	Supero-temporal	15.4	2	7.2	3.58	25	114	6/6	59.00		
33	Male	Left	6/9	Supero-temporal	12.4	4.4	4.7	2.75	20	85	6/7.5	63.00		
34	Female	Left	6/6	Supero-temporal	8.8	3.8	6.3	3.57	15	80	6/7.5	43.00		
35	Male	Right	6/6	Superior	10.1	2.4	5.5	4.5	20	117		no		

Table 1 (continued)

Number	Sex	Eye	Initial visual acuity	Quadrant	Large ultrasound diameter (mm)	Ultrasound height (mm)	Distance of tumor from disc (mm)	Distance of tumor from fovea (mm)	Plaque diameter (mm)	Apex dose (Gy)	Final vision	Follow-up time (months)	Recurrence	Adjunctive TTT
36	Female	Left	6/12	Temporal	15.6	5.4	5	0.97	25	80	6/12	6.00		
37	Male	Right	6/6	Infero-temporal	13.3	3.4	8.9	3.7	20	97	6/6	18.00		
38	Male	Right	6/6	Infero-temporal	9.8	1.4	3.6	1.8	15	152	6/9	no		
39	Male	Left	6/9	Infero-temporal	12.9	4.1	4	1	20	90	6/4.8	60.00	Marginal tumor recurrence (60 months after brachytherapy)	
40	Female	Left	6/9	Infero-temporal	8.6	2.2	2.4	1.7	15	116	6/12	18.00		
41	Male	Right	6/36	Temporal	13.5	5.9	4	0.1	20	80	HM	6.00		
42	Male	Left	6/12	Temporal	8.4	1.9	3.2	0.5	15	124	6/15	53.00		
43	Female	Left	6/6	Inferior	14.2	2.9	7.5	4.1	20	113	6/12	54.00		
44	Female	Right	6/12	Temporal	12.6	5.1	9	4.56	20	80	6/36	9.00		
45	Male	Right	6/6	Inferior	9.8	2.4	7.8	2.53	15	138	6/6	24.00		
46	Female	Left	6/6	Infero-nasal	17.8	4.2	2.7	4.8	25	80	6/9.5	18.00		
47	Female	Left	6/12	Temporal	14.8	1.8	3	0.1	20	159	6/12	42.00		
48	Female	Right	6/6	Supero-temporal	6.4	1.3	6.7	4.92	15	157	6/4.8	42.00		
49	Male	Left	6/6	Infero-temporal	14.4	6	10.5	3.82	20	80	6/6	52.00		
50	Female	Right	6/6	Supero-temporal	9.3	2.8	6.9	3.7	15	103	6/6	32.00		
51	Female	Right	6/12	Temporal	16	3.5	2.4	0.1	25	90	HM	46.00		TTT for macular edema
52	Female	Right	6/6	Supero-temporal	14.4	2.5	2.5	2.78	20	136	6/5	6.00		
53	Male	Right	6/6	Temporal	12.1	3.2	7.7	1	20	93	6/6	49.00	Marginal tumor recurrence (19 months after brachytherapy)	
54	Female	Right	6/12	Temporal	13.1	5.2	8.6	2.96	20	80	6/9	7.00		

visual acuity of 0.5 or better, and 49% had a visual acuity of better than 0.1 five years after brachytherapy [32]. Summanen et al. strongly associated poor visual acuity with tumor to within 1 disc diameter of the optic disc, either alone or in addition to the fovea [33]. In a study of 189 patients by Papageorgiou et al., local tumor control was inadequate in 14% of patients, the risk of this complication being higher if the tumor extended closer than 5 mm of the fovea, possibly because a lower dose of radiation was administered in some of these cases [34].

Only 17% of tumors in our study extended to within 3 mm of the optic disc and this was because we preferred proton beam radiotherapy if we expected brachytherapy to cause optic neuropathy. The exclusion of many patients with juxta-papillary tumors reduces confounding caused by optic neuropathy thereby facilitating interpretation of the results with respect to the tumor's proximity to the fovea. Only 25% of eyes lost vision of 6/12 if the posterior tumor margin was located 3–5 mm from the fovea and our impression is that such visual loss occurred mostly as a result of exudation from to irradiated tumor (i.e., 'toxic tumor syndrome'). Conversely, if the tumor extended closer than 3 mm of the fovea, approximately 75% lost vision of 6/12 or better and we believe that such visual loss occurred mostly because of the high dose of radiation delivered to the fovea (i.e., in addition to visual loss caused by the tumor itself and any exudative complications).

In the 24 patients whose tumor margin was 3–5 mm from the fovea, the six patients with visual loss all had risk factors, which included: pre-existing epiretinal membrane (one patient); diabetes mellitus (three patients); end-stage renal failure (one patient); large tumor size in two patients with an higher risk of macular disease in three patients. In fact, three patients of four with macular complications were diabetic. Two patients had a bulky tumor and hence a greater risk of exudation (i.e., 'toxic tumor syndrome').

In our opinion, an important key to the success of treatment was muscle dis-insertion, which greatly improved exposure thereby facilitating accurate positioning of the plaque. This tends to be avoided if possible by many other surgeons because of concerns about causing diplopia. Such a complication was rare in our series and this was because we took care to measure the knot-to-limbus distances before and after muscle dis-insertion and re-insertion. Another important step in the surgical procedure was to perform trans-scleral transillumination and eliciting the 'sunset sign'. The perforations are located 1 mm from the edge of the template, this distance corresponding to the inactive edge of the plaque. Alternatives to this technique have been described and these include intra-operative ultrasonography and indentation of the eye with the plaque or template [35–38]. We consider these methods to be less reliable and more difficult than trans-scleral transillumination. Some surgeons omit the template, transilluminating the eye so as to silhouette the margins of the radioactive

plaque (Bechrakis, pers. comm.); however, this exposes the staff to higher doses of radiation, especially if it is necessary to re-position the plaque. In non-diabetic patients, a minimum scleral dose of 300–350 Gy was prescribed and this was designed to induce visible choroidal atrophy. When assessing patients post-operatively, such atrophy indicated where the plaque had been positioned in relation to the tumor thereby making it easier to decide which patients could safely be discharged and which ones needed to be reviewed at our specialist center because of an increased risk of local tumor recurrence. Experience over many years has shown that local tumor recurrence is rare even when visible choroidal atrophy does not completely surround the posterior tumor margin. With diabetic patients, an apex dose of 80 Gy is prescribed, irrespective of the dose delivered to sclera, and the tumor is treated with adjunctive transpupillary thermotherapy, which in addition to preventing local tumor recurrence should diminish exudation from the tumor and overlying retina by inducing vascular closure.

Although it might be expected that eccentric positioning of a ruthenium plaque would be followed by an increased incidence of local tumor recurrence, this does not seem to be the case, at least if the surgeon is experienced. An advantage of this strategy is that it results in a wider safety margin anteriorly, where local tumor recurrence may be difficult to detect. Furthermore, because the plaque is further from the optic disc and fovea, it is possible to safely administer higher doses of radiation thereby avoiding central local recurrence.

In conclusion, our study indicates that eccentric plaque placement provides acceptable rates of local tumor control and, if the tumor does not extend closer than 3 mm of fovea, relatively good rates of visual conservation. It must be emphasized that the findings of our study are relevant only to ruthenium plaques. Whether other types of plaque can safely be positioned eccentrically will need to be investigated by other studies.

Conflicts of interest None of the authors has any financial/conflicting interest or funding source to disclose.

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